Shervin Naseri

Senior Machine Learning Engineer | Backend Developer

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PROFESSIONAL SUMMARY

Full Stack Engineer and Technical Lead building resilient microservices, scalable web applications, and AI systems since 2016. Proven expertise in designing Retrieval-Augmented Generation (RAG) pipelines, event-driven architectures, and LLM tool-calling integrations across Python, Node.js, and React. Experienced in delivering document-processing systems for government and enterprise clients.

SKILLS

Backend: Python, Node.js, Java, FastAPI, Express, Microservices, APIs, event-driven systems

Frontend: React, Angular, TypeScript, JavaScript, Next.js, Tailwind CSS

AI / LLM & Retrieval: OpenAI and Anthropic APIs, AWS Bedrock, Vector databases, Chroma, Weaviate, RAG, prompt design, tool calling, embeddings

Data & Architecture: SQL, NoSQL, PostgreSQL, MongoDB, Redis, Prisma ORM, SQLAlchemy

DevOps & Testing: Docker, CI/CD pipelines, GitHub Actions, Pytest, Vitest, observability, Nginx

Certifications: AWS Certified Generative AI Developer - Professional (Amazon Web Services, 2026-03)

EXPERIENCE

AI & Software Engineer

Independent | Ottawa, ON *2025-03 -*

- Architected RAG pipelines and LLM integrations combining vector databases with keyword search for grounded semantic retrieval.
- Built and refactored Node.js and Python services to support structured LLM outputs, schema validation, and tool-calling decoders.
- Engineered asynchronous data processing workflows and optimized PyTorch pipelines for model fine-tuning, embedding generation, and inference.

Graduate Research & Teaching Assistant (part-time)

University of Ottawa | Ottawa, ON *2022-01 - 2024-12*

- Researched the reliability of simulation-based testing for autonomous driving systems; co-authored "Evaluating the Impact of Flaky Simulators on Testing Autonomous Driving Systems" (Empirical Software Engineering; presented at ICSE 2025).

Senior ML Engineer

Pardazesh Etelaat Kian (IraPardaz) | Tehran, Iran *2020-06 - 2021-04*

- Productionized high-accuracy document OCR engines to parse complex layouts, multi-column formats, and structured text blocks.
- Built a perceptual similarity search system utilizing high-dimensional embeddings to index, group, and deduplicate visual and text assets.
- Engineered real-time computer vision and tracking algorithms using YOLOv4 and perspective regression for automated analytics.
- Led technical interviews and mentored junior engineers on testing best practices, validation frameworks, and model deployment.

Co-founder & CTO (Tech Lead)

Nikaat Corp | Tehran, Iran *2016-12 - 2020-04*

- Architected and maintained RESTful microservices and backend APIs for Mint and Phosphor portfolios using Node.js, Express, and MongoDB.
- Designed and deployed Bi-LSTM Seq2Seq deep learning architectures for domain-specific currency exchange rate forecasting.
- Engineered asynchronous backend task queues and predictive pipelines to process complex, multi-modal time-series data.
- Managed CI/CD deployment pipelines and containerized production environments utilizing Docker and Nginx.

Python Developer

Vestaak | Tehran, Iran *2016-06 - 2016-12*

- Developed a content recommendation engine using Python and Scikit-learn to process user interaction data mined via external APIs.

PROJECTS

CodebaseMCP (Multi-Tenant RAG & Architecture Indexing) | https://github.com/shervinemp/CodebaseMCP

- Architected an event-driven Model Context Protocol (MCP) server in Python, integrating Weaviate vector search and AST parsing for multi-tenant code retrieval.
- Integrated Google Gemini APIs with prompt engineering and context grounding to dynamically generate structured technical documentation.

Full-Stack Web Application | https://github.com/shervinemp

- Architected an end-to-end type-safe web platform using Next.js 16, React 19, and tRPC with strict Zod schema validation.
- Engineered database persistence via Prisma ORM with PostgreSQL and high-performance background queues using BullMQ and Redis.

VoxPipe (Real-Time Voice & Agentic Tool-Calling Pipeline) | https://github.com/shervinemp/voxpipe

- Engineered an event-driven voice and reasoning system in Python, orchestrating streaming ONNX models with LLM tool-calling decoders.
- Implemented a conversational memory layer using SQLite WAL mode and FTS5 search indexing for real-time prompt context injection.

EDUCATION

MSc Computer Science

University of Ottawa | *2022-01 - 2025-03*

MSc Software Engineering

Sharif University of Technology | *2018-09 - 2021-10*

BSc Computer Science

Amirkabir University of Technology | *2013-09 - 2018-06*